

Mobile and Pervasive Computing (01CT0716) — Unit 1

Question Bank Answers

PART A: 4 Marks Questions

1. Analyze the use of WSN topologies for smart agriculture

Smart agriculture systems deploy sensor nodes across fields to monitor soil moisture, temperature, humidity, and crop health, then relay this data to a base station connected to the internet. The choice of topology affects coverage, reliability, and cost.

- **Star Topology:** A single coordinator communicates directly with all field sensors. Simple and low-latency, but suitable only for small farms since every sensor must be within direct range of the coordinator.
- **Tree Topology:** Sensors are organized hierarchically, with router nodes extending the coordinator's reach. Useful for medium-sized farms where sensors are spread across multiple sections/plots, each section reporting up through a local router.
- **Mesh Topology:** Nodes relay data through each other, offering the widest coverage and redundancy — ideal for large or irregularly shaped farms where a sensor's data can be rerouted if one node fails (e.g., due to weather or animal damage).

Analysis: For large-scale farms, **mesh topology** is generally the best fit because it provides self-healing paths and extends coverage without needing every node in range of the base station, ensuring reliable delivery of irrigation and climate data despite field obstructions (crops, structures).

2. What is WLAN architecture? List its main components

WLAN (Wireless Local Area Network) architecture, standardized by IEEE 802.11, defines how wireless devices connect, communicate, and interact with a wired backbone network. It provides organized coverage, seamless connectivity, and mobility.

Main Components:

- **Stations (STA):** End-user devices (laptops, smartphones, IoT devices) fitted with a wireless network adapter.
- **Access Point (AP):** Acts as a bridge between wireless stations and the wired network; manages traffic within its coverage area.
- **Basic Service Set (BSS):** A group of stations connected under a single AP — the smallest operational unit of a WLAN. Can be Infrastructure BSS (via AP) or Independent BSS (peer-to-peer, ad hoc).

- **Extended Service Set (ESS):** Multiple BSSs connected together via a Distribution System, offering broader coverage and roaming.
 - **Distribution System (DS):** The wired backbone that links multiple APs/BSSs within an ESS, allowing data to move efficiently across the network.
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3. Compare 1G, 2G, 3G, and 4G technologies based on data rate, switching technique, and key applications

Generation	Data Rate	Switching Technique	Key Applications
1G (1980s)	~2.4 kbps	Circuit switching (analog)	Analog voice calls only
2G (1990s)	9.6–64 kbps	Circuit switching (digital) + packet switching (GPRS)	Digital voice, SMS, basic text messaging, encryption
3G (2000s)	384 kbps–2 Mbps	Packet switching (with circuit switching for legacy voice)	Mobile internet, video calling, multimedia streaming
4G (LTE)	100 Mbps–1 Gbps	All-IP packet switching	HD video streaming, online gaming, VoIP, seamless high-speed data

Each generation improved on the previous by moving from purely analog, circuit-switched voice systems toward fully digital, IP-based, high-throughput data networks capable of supporting multimedia and real-time applications.

4. Compare various Wi-Fi IEEE standards with their specifications

Wi-Fi Generation	IEEE Standard	Frequency Band	Max Data Rate
Wi-Fi 4	802.11n	2.4 / 5 GHz	~600 Mbps
Wi-Fi 5	802.11ac	5 GHz	~3.5 Gbps
Wi-Fi 6	802.11ax	2.4 / 5 GHz	~9.6 Gbps
Wi-Fi 7	802.11be	2.4 / 5 / 6 GHz	~46 Gbps

Older standards like 802.11a/b/g (pre-Wi-Fi 4 naming) operated at lower speeds (up to 54 Mbps) on the 2.4 GHz or 5 GHz ISM bands. Each successive standard increases channel width, uses more efficient modulation (e.g., OFDMA in Wi-Fi 6), and adds multi-user MIMO support, improving speed, capacity, and reduced interference in dense environments.

5. Analyze the differences between connection-oriented and connection-less services in wireless communication

Feature	Connection-Oriented Service	Connection-less Service
Analogy	Telephone system	Postal system
Setup	Requires connection establishment and termination (handshake)	No connection setup or termination
Reliability	Generally reliable, ordered delivery	No guarantee of reliability or order
Packet path	All data follows the same established path	Packets may follow different paths to the destination
Example use	Voice calls, TCP-based sessions	Broadcast messages, UDP-based data, sensor alerts

Analysis: Connection-oriented service suits applications needing guaranteed, ordered delivery (e.g., voice calls in cellular systems), while connection-less service suits lightweight, tolerant applications (e.g., periodic sensor readings in a WSN) where occasional packet loss is acceptable in exchange for lower overhead and faster transmission.

6. Design a Zigbee-based smart agriculture sensor network

Design Outline:

- **Sensor nodes (End Devices/RFD):** Deployed across the field to measure soil moisture, temperature, humidity, and light. Battery-powered, low duty cycle for long life.
- **Zigbee Routers (FFD):** Placed at intervals to relay data from distant end devices toward the coordinator, extending the mesh's range across large fields.
- **Zigbee Coordinator:** Positioned centrally (e.g., at the farmhouse or control shed); collects all sensor data and connects to a gateway/internet link for cloud upload.
- **Topology: Mesh topology** is preferred since it provides multiple redundant paths, ensuring continued operation even if a node fails due to environmental damage.
- **Operation:** Sensor data is sent periodically over the 2.4 GHz band (or 915 MHz for less interference) to the coordinator, which forwards data to a cloud dashboard for farmers to monitor irrigation needs and trigger automated watering.

Justification: Zigbee's low power consumption, low cost, and mesh self-healing capability make it ideal for large, remote agricultural deployments where wired infrastructure is

impractical and battery replacement should be minimized.

7. Differentiate between infrastructure-based and ad-hoc wireless networks with suitable examples

Feature	Infrastructure-Based Network	Ad-Hoc Network
Central Controller	Required (Access Point)	Not required
Setup	Planned	Self-organized/spontaneous
Speed	Faster	Slower
Security	More security options	Fewer security options
Coverage/Scalability	Large, high scalability	Limited, moderate scalability
Example	Wi-Fi LAN, 5G cellular network, campus Wi-Fi	MANET, VANET, Bluetooth, Wi-Fi Direct file sharing

Example scenario: A university campus Wi-Fi where students connect through fixed access points is infrastructure-based, whereas students in a remote field sharing files directly via Bluetooth or Wi-Fi Direct (without any router) form an ad-hoc network.

PART B: 6 Marks Questions

1. Evaluate Wi-Fi and Z-Wave based on range, power consumption, and typical use cases. Which is more suitable for smart home IoT devices?

Feature	Wi-Fi	Z-Wave
Range	~30-50 m indoors	~30-100 m (extended via mesh)
Power Consumption	High (not ideal for battery devices)	Very low (designed for battery-operated sensors)
Data Rate	High (Mbps-Gbps)	Low (~9.6-100 kbps)
Frequency	2.4/5 GHz (congested ISM band)	Sub-1 GHz (908 MHz in US) — less interference
Topology	Star (via AP)	Mesh

Feature	Wi-Fi	Z-Wave
Typical Use Case	Streaming, high-bandwidth devices, smart cameras	Door locks, sensors, light switches, low-power smart home controls

Evaluation: Wi-Fi is better suited for high-bandwidth devices such as smart cameras or streaming speakers, but its high power draw and congestion on the 2.4 GHz band make it less ideal for small battery-operated sensors. **Z-Wave**, operating on a less crowded sub-GHz band with a low-power mesh design, is more efficient and reliable for simple smart home devices like door sensors and light switches.

Conclusion: For smart home IoT devices such as sensors, switches, and locks that need long battery life and reliable low-bandwidth communication, **Z-Wave is more suitable**, while Wi-Fi remains necessary for bandwidth-heavy devices.

2. Apply WLAN architecture to design a wireless network layout for a university campus covering classrooms, labs, and libraries

Design Approach using WLAN Architecture (BSS/ESS/DS):

- **Stations (STA):** Student and faculty laptops, smartphones, and lab computers with wireless adapters.
- **Access Points (APs):** One AP installed per classroom, lab, and library section to form a **Basic Service Set (BSS)** for each area — e.g., separate BSS for each classroom block, one for the library, and dedicated APs in labs (accounting for equipment interference).
- **Extended Service Set (ESS):** All BSSs (classrooms + labs + library) are connected together through the campus's wired backbone, forming a single ESS. This lets a student roam from a classroom to the library while remaining connected without re-authenticating.
- **Distribution System (DS):** A wired Ethernet backbone connects all the APs to the central campus network and internet gateway.

Placement Considerations:

- APs in classrooms placed centrally on the ceiling for even coverage.
- Labs may need APs on the 5 GHz band to reduce interference from lab electronics and support higher throughput for simulation/software work.
- Library APs spaced to cover reading areas and study rooms, using lower-density APs since usage is lighter but spread over a large area.

This layered BSS→ESS→DS structure ensures seamless roaming, centralized management, and scalable coverage across the whole campus.

3. Analyze different WSN network topologies and their suitability for various applications

Topology	Description	Suitability
Bus	All nodes connected to a single shared line	Small, simple networks (e.g., single-room monitoring)
Star	Central master node connects directly to all sensor nodes	Centralized control applications — e.g., small home automation, single-building monitoring
Tree	Hierarchical structure, data flows along branches	Applications needing scalable, structured coverage — e.g., multi-floor building monitoring, agricultural zones
Mesh	Nodes interconnected with multiple paths	Large-scale, critical, or harsh-environment deployments — e.g., industrial monitoring, disaster management, smart agriculture — due to redundancy and robust coverage

Analysis: Star topology offers simplicity and low latency but poor scalability and a single point of failure at the master node. Tree topology balances scalability and structure, suitable for hierarchical deployments like campus-wide environmental monitoring. Mesh topology, while more complex and power-hungry due to multi-hop relaying, provides the highest reliability and coverage, making it the preferred choice for mission-critical or large-area applications such as forest fire detection or precision agriculture.

4. Design a wireless network layout using Wi-Fi which gives coverage around MA102 to MA115, and faculty offices. Specify access point placement and justify coverage

Layout Design:

- **Area:** Rooms MA102–MA115 (a corridor block of 14 rooms) plus adjoining faculty offices.
- **AP Placement:**
 - Place APs at **every 3rd–4th room interval** along the corridor ceiling (e.g., near MA103, MA107, MA111, MA115) to ensure overlapping coverage circles (~30 m radius each) span the whole corridor without dead zones.
 - Faculty offices, being smaller and often walled separately, get **one dedicated AP** placed centrally in the office cluster, using the 5 GHz band to minimize interference with classroom APs and provide higher throughput for faculty work.

- All APs are connected via wired Ethernet (Distribution System) back to the campus network switch, forming a single **Extended Service Set** for seamless roaming between classrooms and offices.

Justification for Placement:

- Spacing APs every 3-4 rooms ensures signal overlap (~10-20%) at BSS boundaries, allowing smooth handoff as users move along the corridor without connectivity drops.
- Concrete walls between rooms attenuate signal, so ceiling-mounted APs (rather than wall-mounted) minimize obstruction and maximize downward coverage into each room.
- Using 5 GHz for faculty offices reduces co-channel interference with the higher-density classroom APs operating on 2.4 GHz, improving overall throughput.
- Centralized DS wiring allows unified network management, consistent SSID, and load balancing across APs.

5. Evaluate the suitability of Wi-Fi and ZigBee for different IoT applications based on power, data rate, and range

Feature	Wi-Fi	ZigBee
Power Consumption	High — unsuitable for long battery life	Very low — designed for battery-run sensors
Data Rate	High (Mbps-Gbps)	Low (20-250 kbps)
Range	~30-50 m (single AP)	~10-100 m per hop, extended via mesh
Network Size	Moderate (dozens of devices per AP)	Very large (up to 65,000 theoretical nodes)
Topology	Star (via AP)	Star, Tree, or Mesh
Typical IoT Use	Smart cameras, video doorbells, smart TVs, high-bandwidth appliances	Smart lighting, HVAC sensors, industrial monitoring, smart agriculture sensors

Evaluation: Wi-Fi is well-suited for IoT devices requiring high throughput or streaming (security cameras, smart displays) but drains battery quickly, making it unfit for small, always-on sensors. ZigBee, with its low power draw, mesh self-healing network, and support for thousands of nodes, is ideal for large sensor deployments such as smart lighting

systems, industrial condition monitoring, or agricultural sensor networks where devices must run for months/years on batteries.

Conclusion: Choose Wi-Fi for bandwidth-intensive, mains-powered IoT devices; choose ZigBee for low-power, large-scale sensor/actuator networks.

6. Compare Wi-Fi and WiMAX

Feature	Wi-Fi (IEEE 802.11)	WiMAX (IEEE 802.16)
Coverage Range	Short range (~30-100 m)	Long range (up to ~50 km)
Data Rate	Up to several Gbps (short range)	Up to 1 Gbps (fixed), ~40 Mbps typical mobile
Frequency	2.4/5/6 GHz (ISM, unlicensed)	2.5, 3.5, 5.8 GHz (licensed and unlicensed)
Mobility	Designed for local, mostly stationary/nomadic use	Supports both fixed (subscriber stations) and mobile stations
Typical Use	Home/office LANs, hotspots, campus networks	Broadband access for rural/underserved areas, metropolitan-area wireless access
Architecture	AP-based BSS/ESS with wired DS backbone	Base station serving subscriber & mobile stations, connected to backbone/internet
Cost/Deployment	Low-cost, easy to deploy for small areas	Higher infrastructure cost but cost-effective vs. wired infrastructure over large areas

Comparison Summary: Wi-Fi is optimized for short-range, high-speed local connectivity (ideal for homes, offices, and campuses), while WiMAX is designed for **metropolitan-scale broadband access**, especially useful in rural or underserved regions lacking wired infrastructure. WiMAX trades some speed and requires licensed spectrum in many bands but delivers dramatically larger coverage per base station compared to Wi-Fi's per-AP range.

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